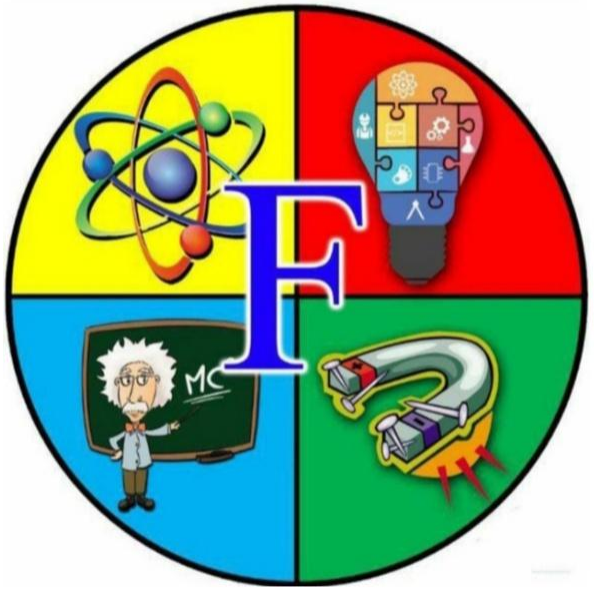
Fizika

9-sinf

Mavzu:

Masalalar yechish.



$$Q_e = \lambda_e m$$

$$Q_b = r_b m$$

$$\varphi = \frac{p}{p_0} \cdot 100\,\% = \frac{\rho}{\rho_0} \cdot 100\,\%$$

23-mashq 1-masala

Qaynash temperaturasida **5 kg** suvni to‘la bug‘ga aylantirish uchun qancha issiqlik miqdori kerak?

Suvning solishtirma bug‘lanish issiqligi

$\mathbf{2,3 \cdot 10^6\ J/kg}$.

Berilgan:

$m = 5\ kg$

$r_b = 2{,}3 \cdot 10^6\ J/kg$

Topish kerak: $Q - ?$

Formula:

$Q = r_b m$

Yechish: $Q = 2{,}3 \cdot 10^6 \frac{J}{kg} \cdot 5\ kg = 11{,}5 \cdot 10^6\ J = 11{,}6\ MJ$

Javob: $Q = 11,6$ **MJ**

23-mashq 3-masala

20℃ temperaturada $\mathbf{4}\,\boldsymbol{m^3}$ havoda **40 g** suv bug‘i bo‘lsa, havoning nisbiy namligi qancha? **20** ℃ temperaturda to‘yingan suv bug‘ining zichligi $\mathbf{17,3}\,\frac{\boldsymbol{g}}{\boldsymbol{m^3}}$.

Berilgan:

$m = 40\ g$

$V = 4\ m^3$

$\rho_0 = 17{,}3\ g/m^3$

Topish kerak: $\varphi - ?$

Formula:

$$\varphi = \frac{\rho}{\rho_0} \cdot 100\ \%$$

$$\rho = \frac{m}{V}$$

$$\varphi = \frac{m}{V\rho_0} \cdot 100\ \%$$

Yechish: $$\varphi = \frac{40\ g}{4\ m^3 \cdot 17{,}3 g/m^3} \cdot 100\ \% \approx 57{,}8\ \%$$

Javob: $\varphi = 57{,}8\ \%$

23-mashq 6-masala

24℃ temperaturada havoning nisbiy namligi **50 %** bo‘lsa, absolyut namlik qancha bo‘ladi? **24** ℃ da to‘yingan suv bug‘ining zichligi $\mathbf{21,8}\,\frac{\boldsymbol{g}}{\boldsymbol{m^3}}$.

Berilgan:

$t = 24^0C$

$\varphi = 50\ \%$

$\rho_0 = 21{,}8\ g/m^3$

Topish kerak: $\rho - ?$

Formula:

$$\varphi = \frac{\rho}{\rho_0} \cdot 100\ \%$$

$$\rho = \frac{\rho_0 \varphi}{100\ \%}$$

Yechish: $$\rho = \frac{21{,}8\frac{g}{m^3} \cdot 50\ \%}{100\ \%} = 10{,}9\ g/m^3$$

Javob: $\boldsymbol{\rho = 10,9\ g/m^3}$

Masala

$10l$ hajmli bir idishda nisbiy namligi **40 %** bo‘lgan havo, **$30\,l$** hajmli boshqa idishda esa temperaturasi xuddi shunday lekin nisbiy namligi **60 %** bo‘lgan havo bor. Idishlar kranli yupqa trubka orqali ulangan.

Kran ochilgandan so‘ng nisbiy namlik qanday bo‘ladi?

Berilgan:

$\varphi_1 = 40\ \%$

$V_1 = 10\ l$

$\varphi_2 = 60\ \%$

$V_2 = 30\ l$

Topish kerak: $\varphi_u - ?$

Formula:

$$V_{1-b} = \frac{V_1 \varphi_1}{100\ \%}$$

$$V_{2-b} = \frac{V_2 \varphi_2}{100\ \%}$$

$$V_{u-b} = V_{1-b} + V_{2-b}$$

$$\varphi_u = \frac{V_{u-b}}{V_u} \cdot 100\ \% \qquad V_u = V_1 + V_2$$

$$\varphi_u = \frac{\frac{V_1 \varphi_1}{100\ \%} + \frac{V_2 \varphi_2}{100\ \%}}{V_1 + V_2} \cdot 100\ \%$$

$$\varphi_u = \frac{V_1 \varphi_1 + V_2 \varphi_2}{V_1 + V_2}$$

Yechish:

$$\varphi_u = \frac{10\ l \cdot 40\ \% + 30\ l \cdot 60\ \%}{10\ l + 30\ l} = 55\ \%$$

Javob: $\varphi_u = 55\ \%$

23-mashq 7-masala

Psixrometrning quruq termometri $\mathbf{24}℃$ ni, nam termometri esa $\mathbf{19}℃$ ni ko‘rsatmoqda. Psixrometrik jadvaldan foydalanib, havoning nisbiy namligini aniqlang.

$$\Delta \boldsymbol{t} = \boldsymbol{t_q} - \boldsymbol{t_n} = \mathbf{24}℃ - \mathbf{19}℃ = \mathbf{5}℃$$

°C	0	1	2	3	4	5	6	7	8
15	100	90	80	71	61	52	44	36	27
16	100	90	81	71	62	54	45	37	30
17	100	90	81	72	64	55	47	39	32
18	100	91	82	73	64	56	48	41	34
19	100	91	82	74	65	58	50	43	35
20	100	91	83	74	66	59	51	44	37
21	100	91	83	75	67	60	52	46	39
22	100	92	83	76	68	61	54	47	40
23	100	92	84	76	69	61	55	48	42
24	100	92	84	77	69	62	56	49	43
25	100	92	84	77	70	63	57	50	44

Mustaqil bajarish uchun topshiriqlar

1. Massasi 50 g bo‘lgan suv bug‘i kondensat-siyalanganda qancha issiqlik ajraladi? (23-mashq 2-masala)

2. $20℃$ temperaturada havodagi suv bug‘ining bosimi 1,54 kPa ga teng. Agar $20℃$ temperaturada to‘yingan bug‘ bosimi 2,43 kPa bo‘lsa, nisbiy namlikni aniqlang. (23-mashq 4-masala)

3. $20℃$ temperaturada havodagi suv bug‘ining zichligi 17 g/m^3 ni tashkil qiladi. Agar havoning absolyut namligi 11 $\frac{g}{m^3}$ bo‘lsa, havoning nisbiy namligini toping. (23-mashq 5-masala)